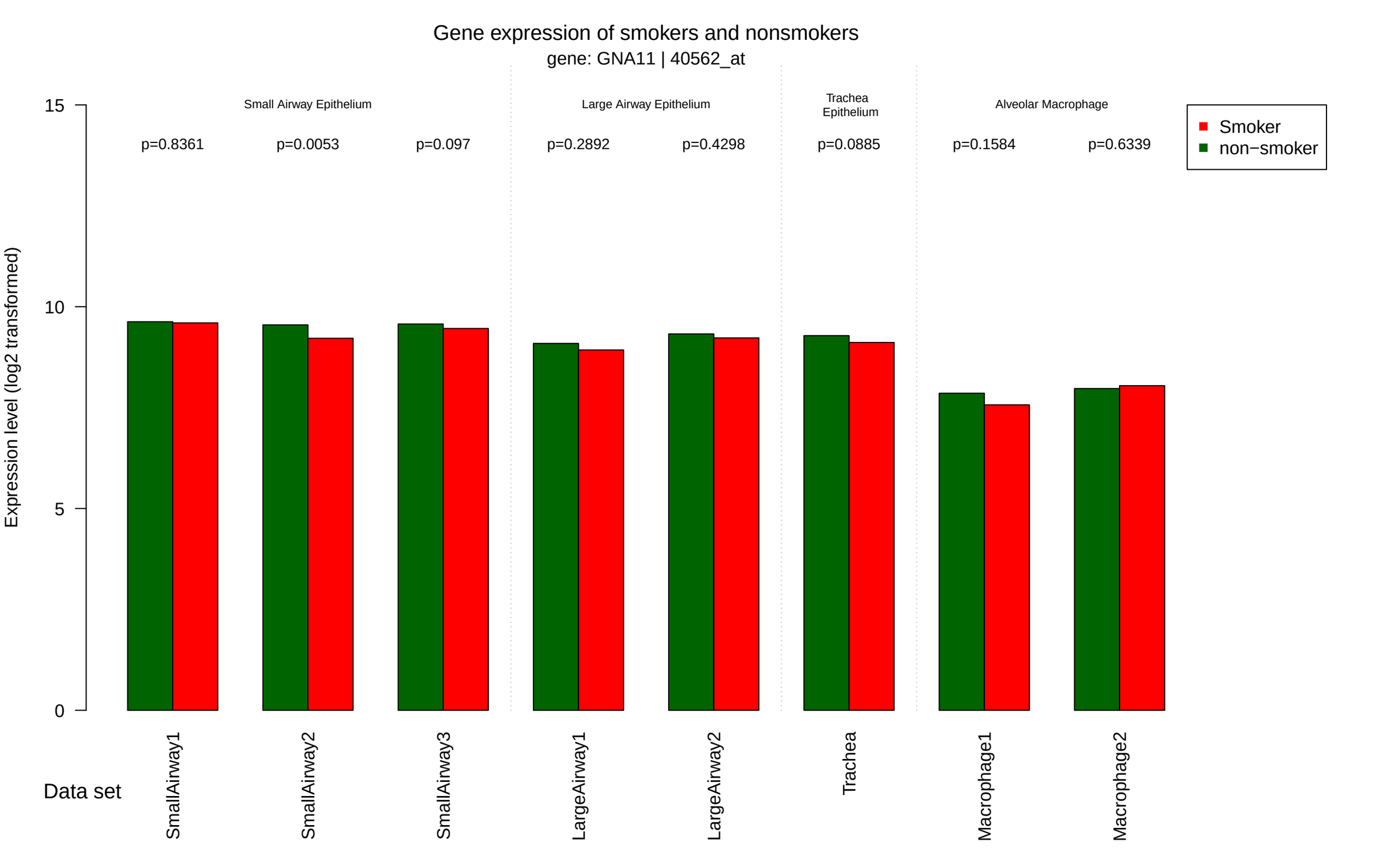
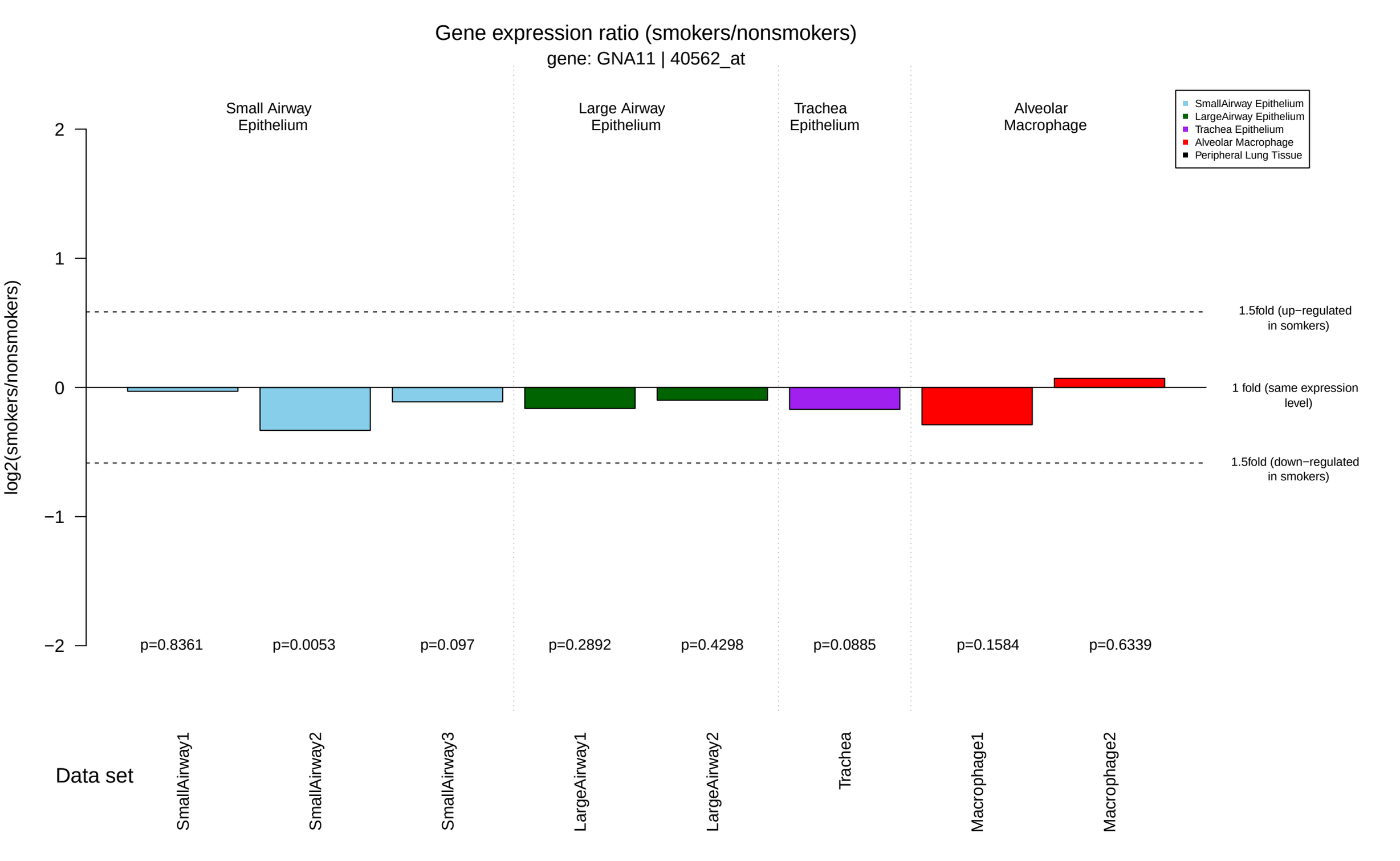
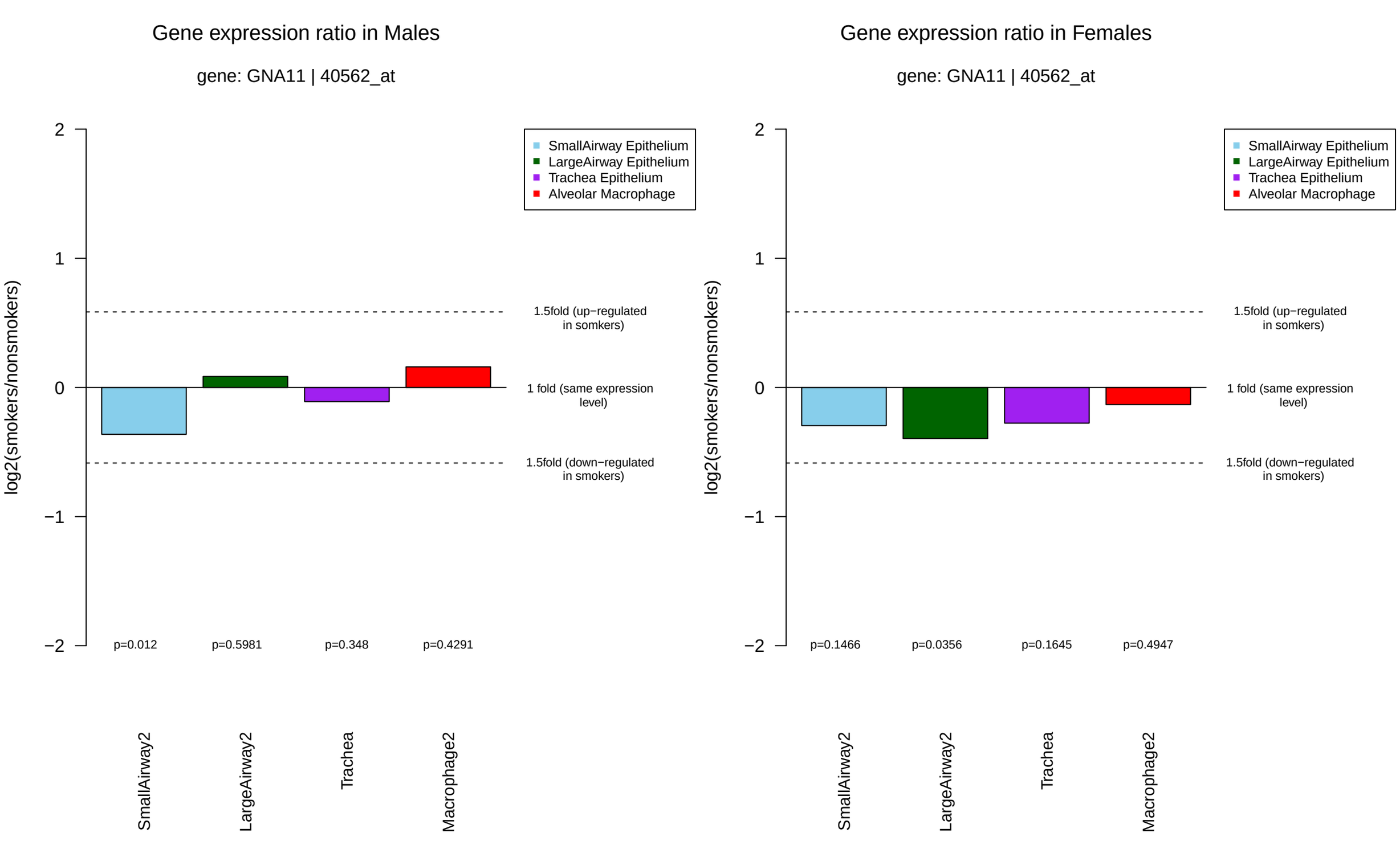
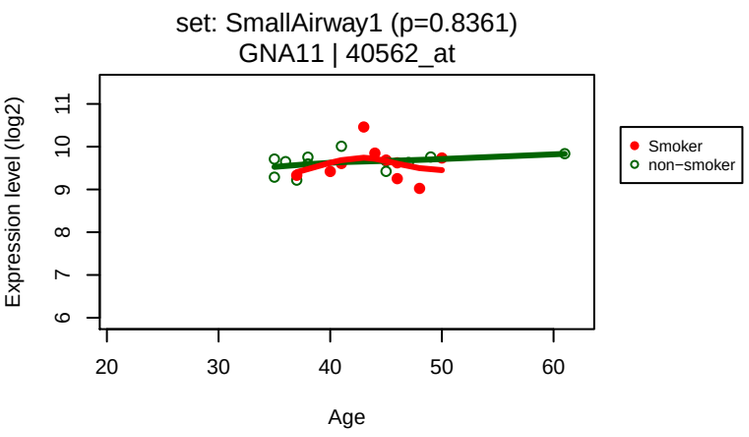
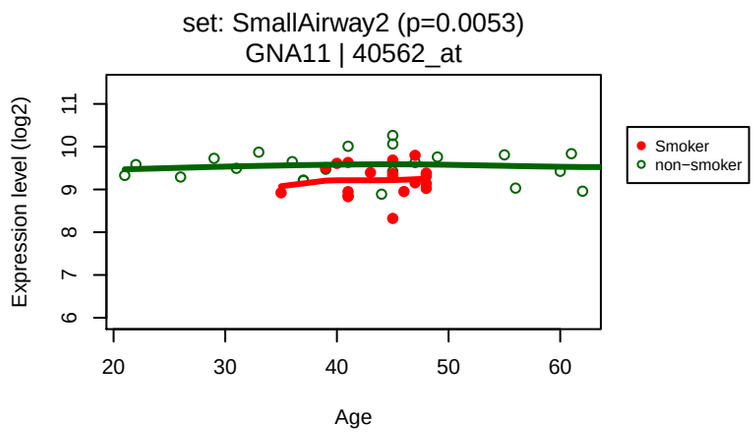
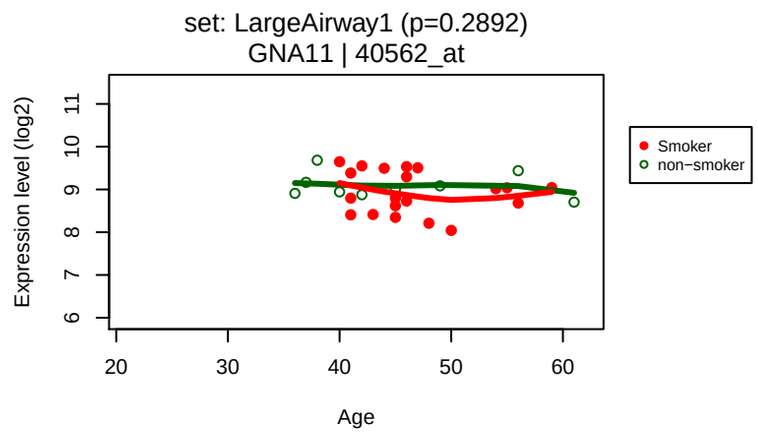

Gene expression of smokers and nonsmokers
gene: GNA11 | 40562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8361
p=0.0053
p=0.097
p=0.2892
p=0.4298
p=0.0885
p=0.1584
p=0.6339
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

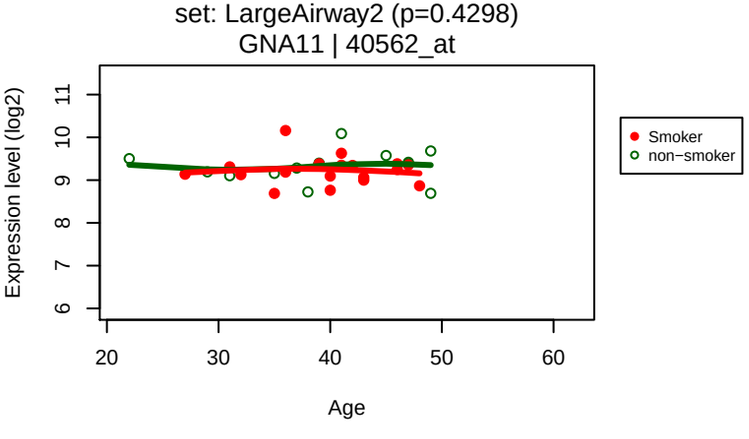

Gene expression ratio (smokers/nonsmokers)
gene: GNA11 | 40562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8361
p=0.0053
p=0.097
p=0.2892
p=0.4298
p=0.0885
p=0.1584
p=0.6339
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

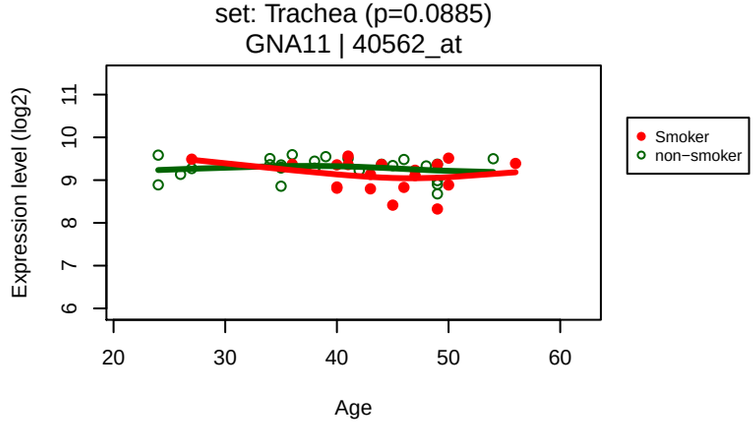

Gene expression ratio in Males
gene: GNA11 | 40562_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.012
p=0.5981
p=0.348
p=0.4291
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNA11 | 40562_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1466
p=0.0356
p=0.1645
p=0.4947
SmallAirway2
LargeAirway2
Trachea
Macrophage2

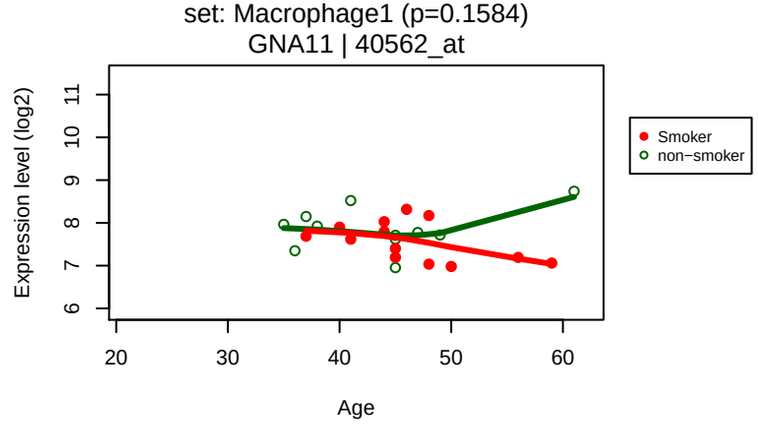

set: SmallAirway1 (p=0.8361)
GNA11 | 40562_at
Expression level (log2)
Age
Smoker
non−smoker

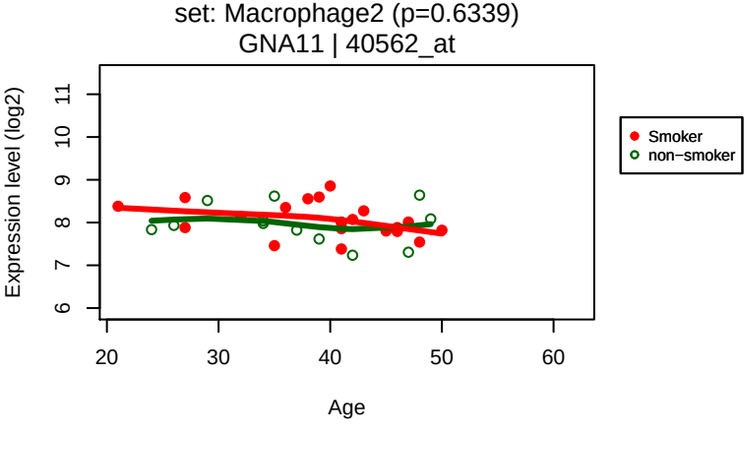

set: SmallAirway2 (p=0.0053)
GNA11 | 40562_at
Expression level (log2)
Age
Smoker
non−smoker

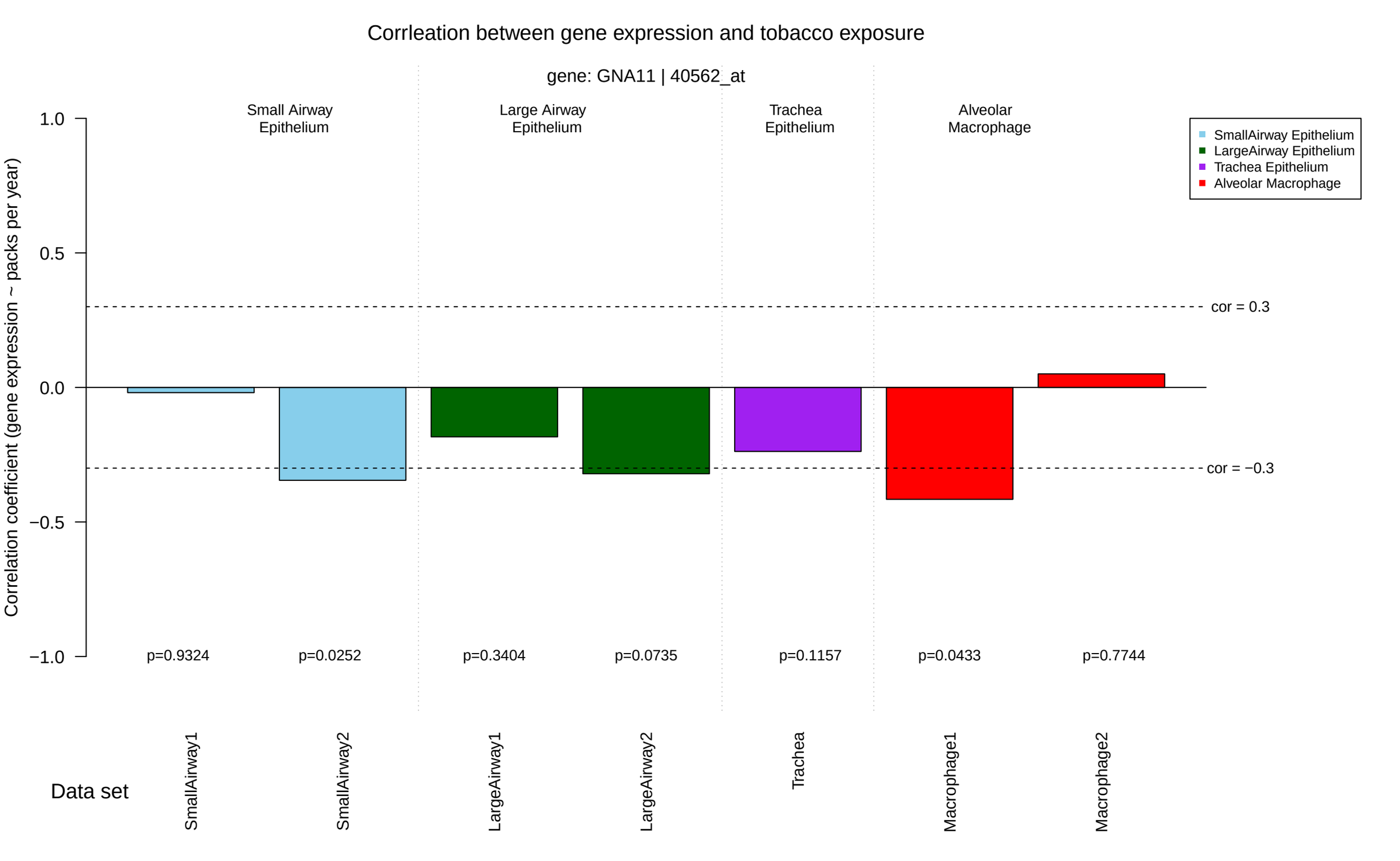

set: LargeAirway1 (p=0.2892)
GNA11 | 40562_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4298)
GNA11 | 40562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0885)
GNA11 | 40562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1584)
GNA11 | 40562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6339)
GNA11 | 40562_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNA11 | 40562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9324
p=0.0252
p=0.3404
p=0.0735
p=0.1157
p=0.0433
p=0.7744
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2